

Lab 3: Component Modelling & Architectural Pattern Selection

Architectural Justification — Alumni Mentorship & Mock Interview Platform (Problem Statement #06)

Chhavi S Wadhwa | SRN: PES1UG24CS132 | Section: C | Project Key: AMF

Architecture Selection: We chose the Layered (N-Tier) Architecture for the Alumni Mentorship & Mock Interview Platform System.

The system is organised into three layers — a Presentation Layer (User Interface Component), a Business/Application Layer (Auth & Profile Manager, Mentor Matching & Booking Manager, Notification & Meeting Link Service, Scorecard Manager), and a Data Layer (Database Component), with the Notification service depending on an external Video-Conferencing API. This mirrors the accompanying UML Component Diagram.

Reason 1 — Separation of concerns aligned to the backlog

The Lab 2 backlog splits the product into three independent epics — Authentication (Epic 1), Mentor Discovery & Booking (Epic 2), and Session Management & Evaluation (Epic 3). A layered architecture lets each epic map to its own Business-layer component (Auth & Profile Manager, Mentor Matching & Booking Manager, Scorecard Manager) that sits behind a common Presentation layer and above a shared Data layer. Teams can therefore develop and sprint on one epic's component without touching the UI or database code owned by another, which is exactly the isolation the sprint-based delivery model (18 story points/week velocity, Lab 2) depends on.

Reason 2 — Centralised consistency for booking logic

Double-booking prevention (FR-001, FR-002, and Alternate Flow 5a of UC-03) requires every calendar read/write to be checked against one authoritative source of truth. A layered design funnels all such operations through a single Business-layer component (Mentor Matching & Booking Manager) to a single Data-layer component (Database), avoiding the eventual-consistency and distributed-transaction problems a microservices split would introduce for this concurrency-sensitive operation, at a scale (a single campus platform) that does not need independent service scaling.

Security Advantage

NFR-001 requires profile data and scorecards to follow privacy guidelines and support soft-deletion within 24 hours. In the layered design, the Presentation layer has no direct interface to the Database Component — every access to persisted data must pass through a Business-layer component's required interface first. This forces authentication, authorisation, input validation, and soft-deletion logic to be enforced in one place instead of being duplicated (or accidentally skipped) in the UI, shrinking the attack surface for unauthorised data access.

Performance Benefit

NFR-002 requires the mentor-matching endpoint to respond within 3 seconds for 95% of requests at up to 500 concurrent users. Because the layered architecture keeps the Presentation → Business → Data call path within tightly-coupled, in-process/same-deployment calls rather than routing through independently deployed services over a network, it removes the inter-service network hops and serialization overhead that a microservices architecture would add, making the 3-second SLA easier to meet without extra infrastructure such as a service mesh or distributed tracing.